Data Management Plan - Group 21

Project Title: AI-Powered Credit Default Risk Prediction Using Machine Learning and Neural Network Techniques

1. Overview of the Datasets:

This project uses two publicly available financial datasets for credit risk prediction and classification tasks.

Dataset 1: Default of Credit Card Clients

- Source: UCI Machine Learning Repository and Kaggle
- Instances: 30,000
- Features: 23 input variables and 1 target variable
- Task: Classification
- Domain: Financial Risk Analysis

This dataset contains demographic information, payment history, bill statements, and default payment status of credit card clients in Taiwan. The aim is to predict whether a customer will default on their credit card payment.

Official Sources:

- UCI Default of Credit Card Clients Dataset
- Kaggle Default of Credit Card Clients Dataset

Dataset 2: German Credit Data

- Source: UCI Machine Learning Repository and Kaggle
- Instances: 1,000
- Features: 20 attributes
- Task: Classification
- Domain: Banking and Credit Risk

This dataset classifies applicants as good or bad credit risks using financial and demographic information. Both datasets are widely used for machine learning and financial risk analysis research.

Official Sources:

- UCI German Credit Dataset
- Kaggle German Credit Dataset

2. Data Collection:

The datasets were obtained from publicly available academic repositories:

Dataset	Source	Original Provider
Default of Credit Card Clients	UCI / Kaggle	Yeh & Lien (2009)
German Credit Data	UCI / Kaggle	Prof. Hans Hofmann

The datasets are downloaded directly from the UCI Machine Learning Repository and Kaggle. These sources provide data for educational and research purposes.

3. Metadata:

Metadata documentation is available through the official dataset descriptions provided by UCI and Kaggle.

Default of Credit Card Clients:	German Credit Data:
Demographic information	Credit history
Credit limit	Loan duration
Payment history	Employment status
Bill statements	Savings information
Previous payment amounts	Property information
Default payment status	Credit risk classification

4. Document Control and GitHub Usage:

Version control for the project will be maintained using GitHub.

Repository: https://github.com/REKNOS/AI-Powered-Credit-Default-Risk-Prediction-System

GitHub will be used to maintain:

- Version history
- Change tracking
- Collaboration records
- Backup copies of project files

5. README File:

A README file will also be included to explain project objectives, dataset information, installation requirements, libraries used, project structure, and instructions for running the code.

Project Objectives: To develop a machine learning model for predicting customer credit default risk.

Dataset Information: The project uses publicly available financial and credit datasets collected from Kaggle and UCI repositories.

Installation Requirements: Python, Jupyter Notebook, and required machine learning libraries must be installed before running the project.

Python Libraries Used: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and XGBoost are used for data analysis and model development.

Instructions to Run the Code: Open the Jupyter Notebook and execute the cells sequentially after installing the required libraries.

Machine Learning Models Implemented: Logistic Regression, Random Forest, Decision Tree, and XGBoost models are implemented for prediction.

Folder Structure: The project folder contains datasets, notebooks, source code, model files, results, and documentation.

Results Summary: The implemented models successfully analysed credit risk and achieved effective prediction accuracy for loan default classification.

6. Security and Storage:

Storage:
Project files and datasets will be securely stored:

- On a password-protected personal computer
- In private GitHub repositories during development
- On cloud backup storage such as Google Drive or OneDrive

Security Measures:

- Only authorised users will have repository access
- Regular backups will be maintained
- No sensitive personal identifiers are included in the datasets
- Data will not be modified beyond preprocessing requirements

Since the datasets are publicly available and anonymised, the risk of data exposure is minimal.

7. Ethical Statement:

1. Does the data come under GDPR requirements?

The datasets are publicly available and anonymised, meaning they do not contain directly identifiable personal information such as names, phone numbers, addresses, or email addresses. Therefore, the project is unlikely to fall under strict GDPR personal data requirements. However, ethical handling and secure storage practices will still be followed.

2. Does the project conform to UH ethical policies?

Yes. The project conforms to University ethical policies because:

- The datasets are publicly available for research purposes
- No human participants are directly involved
- No personal or confidential information is collected
- The research focuses on predictive modelling and analysis only

3. Do you have permission to use the data for your proposed research project?

Yes. The datasets are openly available through UCI Machine Learning Repository and Kaggle for academic and research use. Proper citations and acknowledgements will be provided in the final report.

4. Are you assured that the data was collected in an ethical manner?

Based on the documentation provided by UCI and Kaggle, the datasets were collected and published by recognised academic researchers and institutions for research purposes. The datasets have been widely used in academic publications and machine learning studies. Therefore, there is reasonable assurance that the data was collected ethically and responsibly.